LONGA MAIMBO

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PROFESSIONAL SUMMARY

MSc Mechatronics Engineering Student at the University of Cape Town specializing in computer vision and artificial intelligence. I integrate computer vision and deep learning with CAD modeling, electronics design, and firmware configuration to deploy autonomous systems. Seeking roles in mechatronics, embedded systems, industrial automation, and AI engineering to bridge advanced AI research and real-world applications.

SKILLS

- **Programming & Frameworks:** Python, C/C++, MATLAB, Simulink, PyTorch, ROS2, Linux, Open3D, Gazebo, GTSAM, PX4.
- **Engineering & Design:** Embedded Design (KiCAD), SolidWorks.
- **Professional:** Technical Communication, Team Leadership, Microsoft Office Suite.

EDUCATION

MSc Eng Mechatronics | University of Cape Town
2025 – Present

BSc Eng (Hons) Mechatronics | University of Cape Town
2021 – 2024 — Completed, Graduated with Honours

A-Levels | Springfields School Of Education
2019 – 2020 — Completed

PROJECTS

3D Reconstruction for Autonomous Drone Navigation (Master's Project) | University of Cape Town

- **Role & Impact:** Developing an autonomous navigation and real-time 3D reconstruction system for low-cost indoor Micro Aerial Vehicles (MAVs), integrating custom Visual Inertial Odometry (VIO), neural depth estimation, volumetric mapping, and predictive flight control.
- **Skills & Tools:** PyTorch, ROS2, DINOv2, GTSAM, CUDA, MAVLink, Ardupilot, Computer Vision, Hardware Synchronization.

End-to-End ECG Digitisation with Soft Segmentation & BiLSTM Signal Reconstruction | Maimbo & McIntosh (2026)

- **Role & Impact:** Coded the classical CV baseline and BiLSTM model for a fully automated three-stage pipeline overcoming noise in paper ECG records. The paper was accepted into the Computing in Cardiology conference in Madrid and is set to be published later this year.
- **Skills & Tools:** PyTorch, OpenCV, GPU Acceleration, Image Rectification, Sequential Modeling.

Monophonic Piano Music Transcription Using Machine Learning (Undergraduate Thesis) | University of Cape Town (2024)

- **Role & Impact:** Developed a pipeline for audio analysis and data extraction to train and test deep learning models for piano transcription in a comparative study.
- **Skills & Tools:** TensorFlow, Keras, Deep Learning, Python, High Performance Computing.

Encouraging Engineering Through Hardware Hackathons | InventAfrika (2025)

- **Role & Impact:** Facilitated and organized hackathons across Cape Town, Stellenbosch, and Windhoek to promote hardware design and 3D printing.
- **Skills & Tools:** Arduino, STM32, IoT, Technical Documentation, Networking.

Oil Spill Detection Using Synthetic Aperture Radar Satellites | University of Cape Town (2025)

- **Role & Impact:** Built a pipeline to pull satellite data and developed a radar band mathematics algorithm to optimize oil spill visibility.
- **Skills & Tools:** Openeo, Python, Jupyter Notebooks, SAR Imaging, Pandas.

Predictive Insights Graduate Hiring Process Predictor Hackathon | Predictive Insights (2025)

- **Role & Impact:** Developed a predictive analytics pipeline to analyse candidate application datasets and forecast progression through the multi-stage graduate hiring process, securing first place in a competitive hackathon hosted with Leaply, Alphawave Group, and the UCT Developer Society.
- **Skills & Tools:** Python, Pandas, NumPy, Predictive Modeling, Exploratory Data Analysis, Statistical Evaluation

WORK EXPERIENCE

New Venture Planning Practice & Professionalism Teaching Assistant | University of Cape Town

July 2026 – Present

- Acted as first point of contact for administrative and course-related student queries.
- Automated administrative processes using code to ensure clean, accessible and easily repeatable student data structures.

Mechatronics Teaching Assistant | University of Cape Town

February 2026 – June 2026

- Acted as first point of contact for administrative and course-related student queries.
- Facilitated laboratory sessions and graded technical projects focusing on systems modeling and control design.

Contract AI Engineer | Muna Africa

July 2025 – October 2025

- Developed a multi-modal machine learning pricing model to automatically categorize and value clothing inventory.
- Integrated computer vision and NLP techniques to process product images and text for accurate price predictions.

Power Electronics Lab Intern | University of Cape Town

June 2023 – August 2023

- Analyzed and proposed design fixes for a doubly fed induction panel.
- Implemented design fixes and successfully deployed the induction panel.

REFERENCES

- Stacey Shield: Course Convener, University of Cape Town - stacey.shield@uct.ac.za
- Tshepo Bhengu: Owner, Muna Africa - tshepo@muna.africa
- Fred Nicolls: Research Advisor - Fred.Nicolls@uct.ac.za
- Ekaterina Rzyankina: Course Convener, University of Cape Town - ekaterina.rzyankina@uct.ac.za